Elia Zanini – UNSUPERVISED LEARNING

Abstract

With an emphasis on unsolved murders, this unsupervised learning project investigates trends in a large dataset of homicide cases in the United States. The aims of using clustering techniques are to find hidden structures in the data, identify common traits of victims and crimes and detect potentially suspicious patterns.

The project aims to offer insights that could aid investigative journalism or law enforcement efforts by classifying cases that are similar. Tens of thousands of recorded homicides with comprehensive details like year, location, weapon, victim demographics, and whether the case has been solved are included in the dataset, which comes from the Murder Accountability Project. Exploratory data analysis, dimensionality reduction, and clustering algorithms are used to identify and analyse underlying patterns in the cases.

1. Introduction

Homicide is one of the most serious and socially impactful crimes and understanding its trends is crucial for both public safety and criminal justice. Thousands of murder cases in the U.S. are unsolved despite major advancements in forensic technology and investigative methods. Finding recurring characteristics in murders may help lead future investigations.

The aim of this project is to use unsupervised learning techniques to examine homicide data. By using this method, I hope to identify any common trends that might have been ignored and identify clusters that match specific victims, weapons or crime settings.

The study makes use of information from the Murder Accountability Project, a program that gathers and standardizes public homicide statistics from all around the United States. This dataset, which contains tens of thousands of records, presents a strong basis for identifying statistical patterns. The methods used to prepare and evaluate this data in order to uncover insightful groupings are investigated in the section that follows.

2. Dataset Overview and Objective

The dataset has more than 600,000 rows of homicide records reported by police departments across the United States: however, for this study, we filtered the dataset to focus on relevant and usable cases: particularly solved cases with consistent information.
The first thing I did was to clean data, keeping just registered information (no NA), solved homicides and filtering starting from year 1990. Both for cleaning and ri-dimension the too bigger dataset.

Key variables (before cleaning) include:

- Year: year of the homicide
- State, Agency, Source: geographic and administrative information
- Crime Type: whether it is a murder or manslaughter
- Victim Sex, Victim Age, Victim Race, Weapon
- Relationship with the victim

The following sections describe the data cleaning, feature selection, transformation processes, and the insights derived from clustering results.

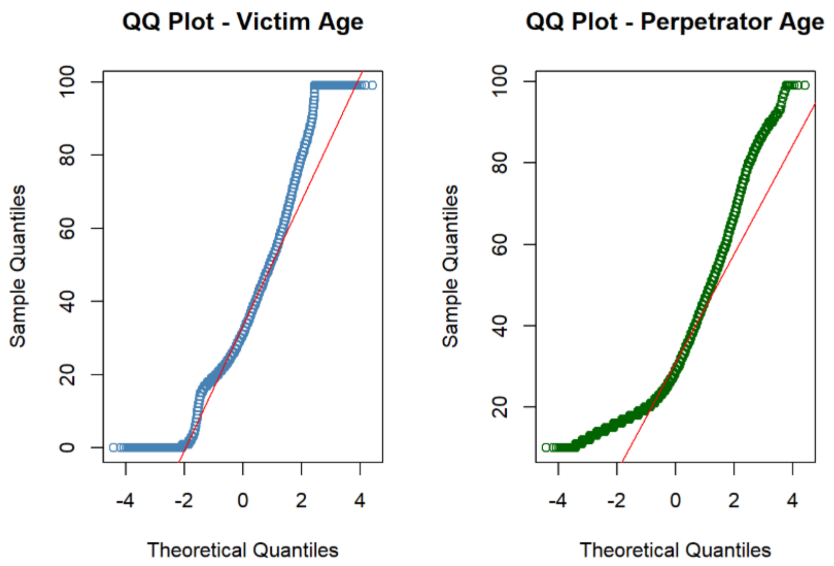
```
## 'data.frame':    102724 obs. of  9 variables:
##  $ Crime.Type      : chr  "Murder or Manslaughter" "Manslaughter by Negligence" "Manslaughter by
e" "Murder or Manslaughter" ...
##  $ Victim.Sex      : chr  "Female" "Male" "Male" "Male" ...
##  $ Victim.Age      : int  25 1 0 32 30 2 21 33 3 3 ...
##  $ Victim.Race     : chr  "Black" "Asian/Pacific Islander" "Asian/Pacific Islander" "White" ...
##  $ Perpetrator.Sex : chr  "Male" "Male" "Female" "Male" ...
##  $ Perpetrator.Age : int  28 12 39 39 20 20 21 20 19 23 ...
##  $ Perpetrator.Race: chr  "Black" "Asian/Pacific Islander" "Asian/Pacific Islander" "White" ...
##  $ Relationship    : chr  "Wife" "Family" "Acquaintance" "Stranger" ...
##  $ Weapon          : chr  "Handgun" "Blunt Object" "Blunt Object" "Rifle" ...
```

3. Feature Transformation and Variable Handling

Each variable was carefully analysed and processed to ensure its suitability for clustering. Since unsupervised learning is highly sensitive to input features, I applied customized transformations based on the type and distribution of each variable.

- *Victim Age / perpetrator Age*

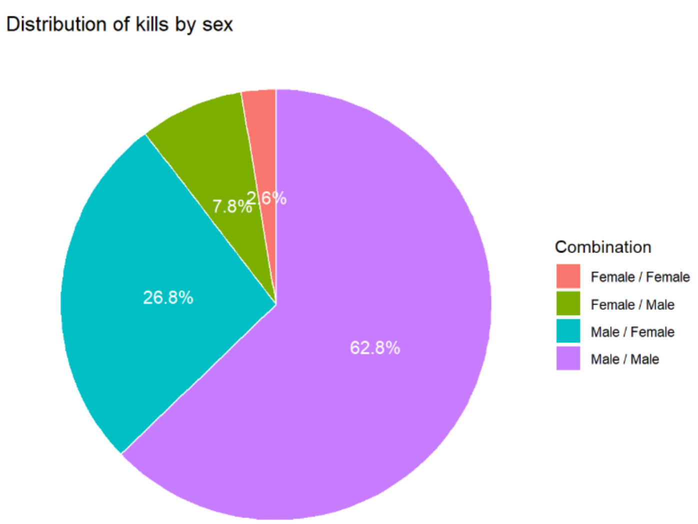
The variables were already retained as numeric. Outliers (es. ages above 100 or below 1) were removed as they were likely due to input errors.

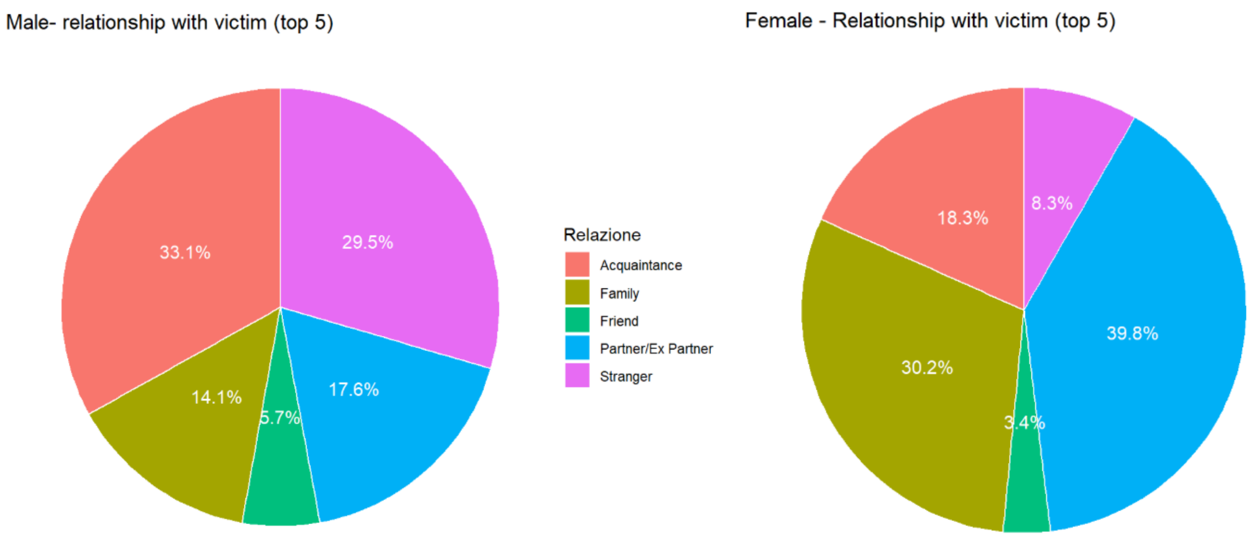


- *Victim Sex / perpetrator Sex*

This binary variable ("Male"/"Female") was encoded as 0 and 1. Unknown or missing values were removed, as they were not numerous and could bias the clustering process.

The goal of the whole study was to find patterns in homicide cases, so start understanding the dataset from some statistics about homicides per sex have been both useful and interesting.





- *Victim Race / Perpetrator Race*

Originally a categorical variable with values such as "White", "Black", "Asian", "Hispanic", and others. After small research I found the best (and lighter) way to keep the information about both the columns in just one dummy:

1: Victim and Killer have same Ethnicity, 0: No

- *Weapon:*

Rather than keeping all weapon types as separate dummies, I created a single binary variable indicating whether the weapon used was a **firearm** or not. This decision was made to reduce dimensionality and focus on the most relevant contrast in homicide profiling: all types of firearms ("Handgun", "Rifle", "Shotgun", etc.) were grouped under the "Firearm" label (coded as 1), while all other weapon types were grouped as 0.

-Crime Type:

The Crime.Type variable was also converted into a binary dummy: 1 if the case was categorized as "Murder", and 0 otherwise (e.g., manslaughter or unspecified types); this transformation helps isolate the most severe crimes for clustering purposes.

-Relationship with victim

The victim–offender relationship was recoded into a dummy variable called Rel_NotClose, which takes value 1 when the offender was a stranger, acquaintance, employer, or employee, and 0 otherwise: this simplification aims to distinguish between homicides committed by close relations versus those by more distant or unknown individuals.

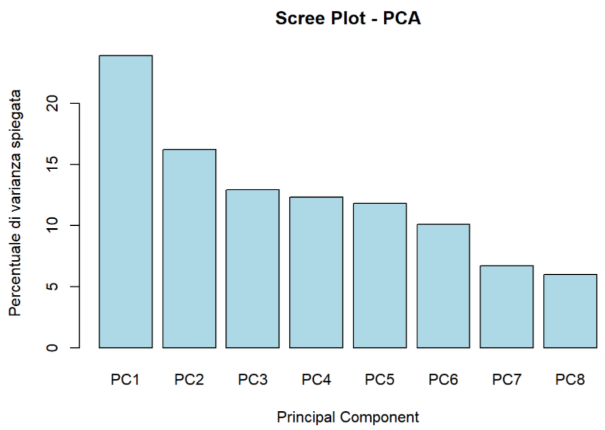
##check finale

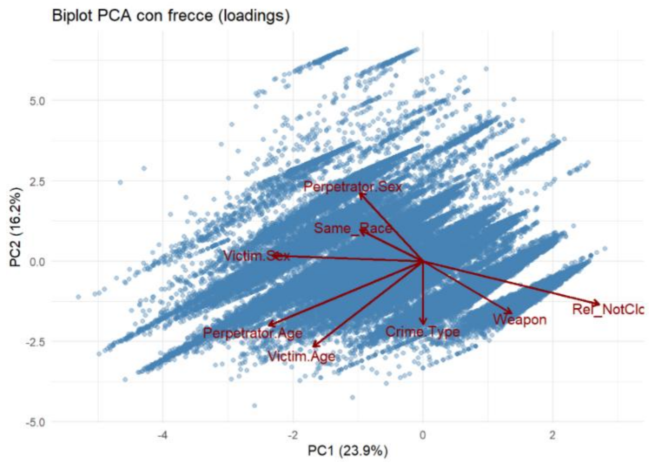
```
str(df)
```

```
## 'data.frame':    102724 obs. of  8 variables:
##  $ Crime.Type      : num  1 0 0 1 1 1 1 1 0 0 ...
##  $ Victim.Sex      : num  1 0 0 0 0 0 0 1 0 0 ...
##  $ Victim.Age      : int  25 1 0 32 30 2 21 33 3 3 ...
##  $ Perpetrator.Sex: num  0 0 1 0 0 0 0 0 0 1 ...
##  $ Perpetrator.Age: int  28 12 39 39 20 20 21 20 19 23 ...
##  $ Weapon          : num  1 0 0 1 1 0 1 0 0 0 ...
##  $ Rel_NotClose    : num  0 0 1 1 1 1 1 1 1 0 ...
##  $ Same_Race       : num  1 1 1 1 1 1 0 0 0 1 ...
##  - attr(*, "na.action")= 'omit' Named int 232235
##   ..- attr(*, "names")= chr "232235"
```

4. Dimensionality Reduction: PCA

At the end of the transformation phase, the dataset was standardized using **z-score normalization** to ensure that all numerical features contributed equally to the distance metrics used by the clustering algorithm. The explained variance plot, also known as a scree plot, indicated that the first two principal components (PC1 and PC2) collectively accounted for an important amount of the variance, sufficient to offer insightful two-dimensional visualization.





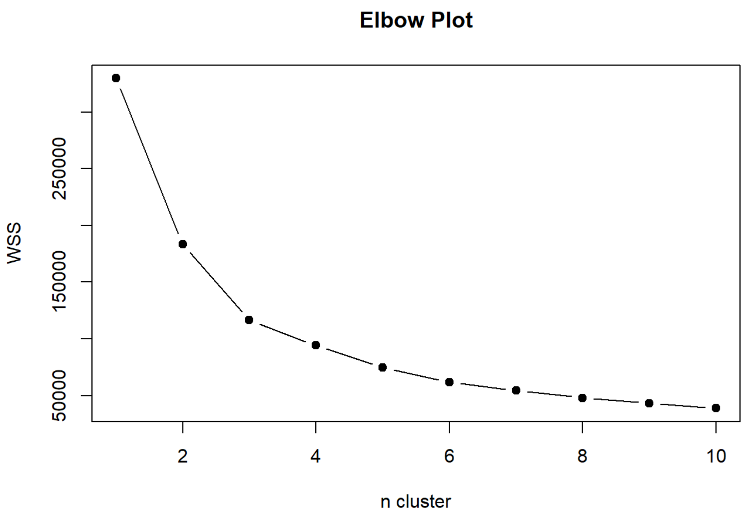
In addition to decreasing the dimensionality for visualization, PCA helped prepare the data ready for clustering by avoiding noise and redundancy. The distribution of observations in the condensed space and the contributions of variables to each component were visually represented by the biplot (see below).

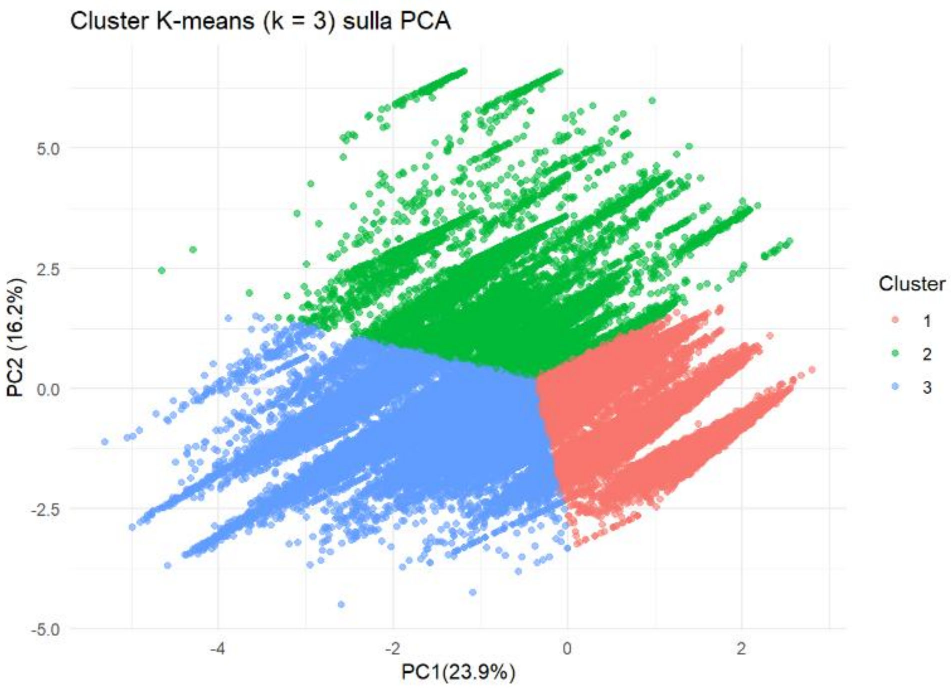
5. Clustering Analysis

After dimensionality reduction with PCA, clustering was performed using the K-means algorithm. The input was the PCA-reduced dataset (first two principal components) ensuring that clustering focused on the main directions of variance in the data.

To determine the optimal number of clusters, the **elbow method** was applied and the total within-cluster sum of squares (WSS) was plotted against a range of possible cluster counts; the elbow point appeared at **k = 3** indicating a natural grouping into three distinct clusters.

Then the K-means algorithm was applied, and each observation was assigned to one of the three clusters.





6. Cluster Interpretation

After studying the mean values for each cluster, I found the pattern for the three cases.

Cluster 1 – Gang Violence / Criminal Conflicts

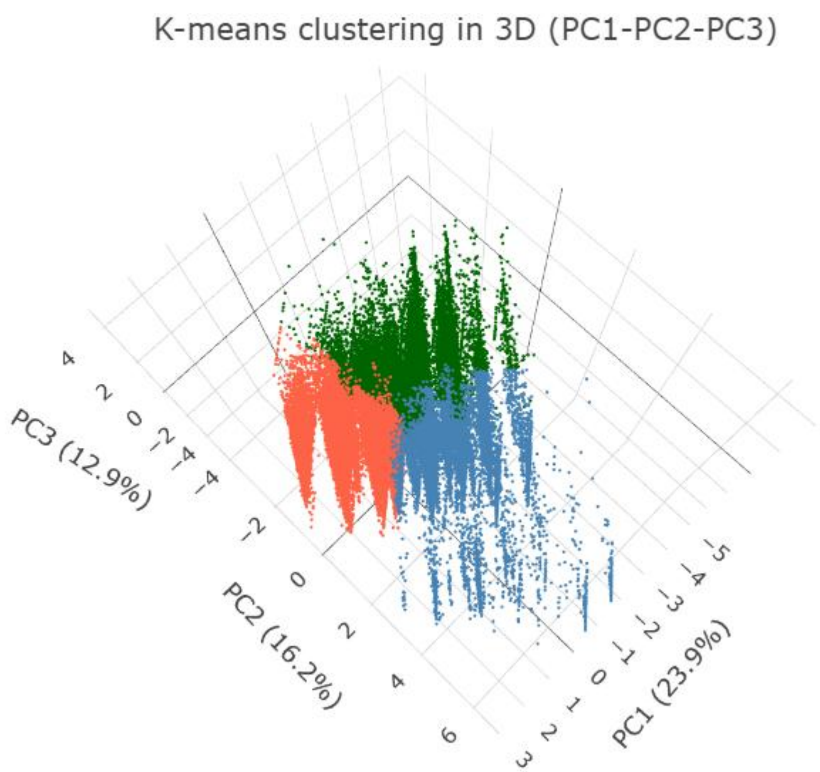
This cluster mainly includes young male victims, often killed with firearms and in cases involving non-close or unknown relationships. The patterns suggest typical features of gang-related activity, street violence, or criminal rivalries, where the victim and perpetrator are not personally connected.

Cluster 2 – Family-Related Violence

Cluster 2 is characterized by very young victims (average around 21 years old), a surprisingly high share of female perpetrators, and a strong prevalence of close relationships between victim and perpetrator indicating family or household-related violence. These cases may involve children or young adults killed by parents or relatives, reflecting tragic scenarios within domestic environments.

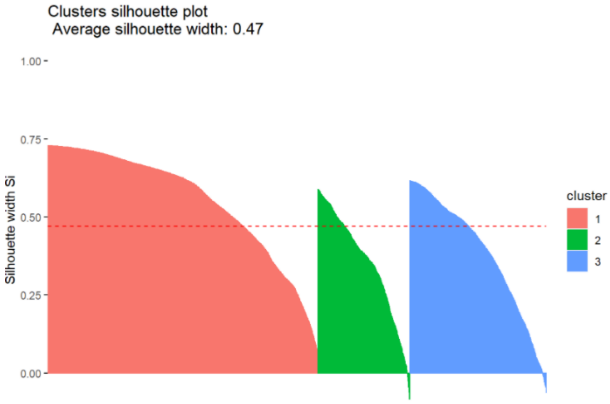
Cluster 3 – Domestic Violence / Partner Homicide

This group involves older victims and perpetrators (average age between 45–50 years), with a majority of female victims and mostly male perpetrators. The relationship in these cases is often "important", suggesting patterns of intimate partner violence, including spousal or ex-partner homicides: one of the most discussed scenario in the last years.



7. Silhouette Analysis

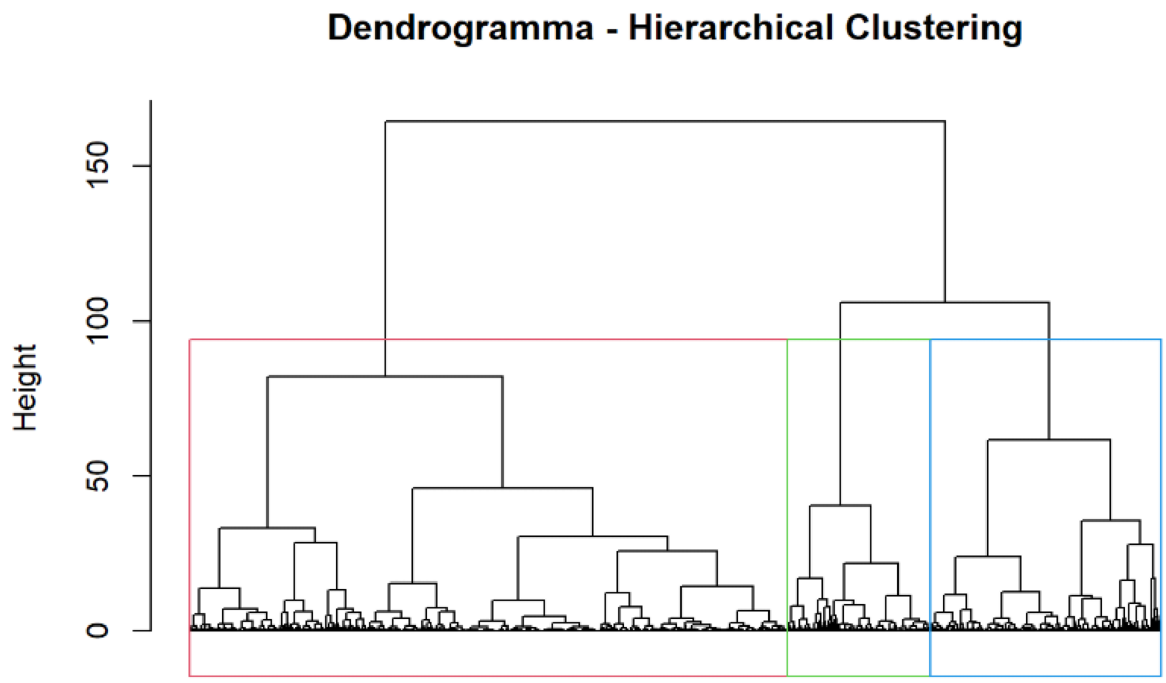
To assess the validity of the clustering results, I computed the **Silhouette Score** for the 3 clusters; this metric evaluates how well each observation fits within its assigned cluster compared to others: not only finding the average Silhouette width (0.47), but also identifying outliers based on their negative Silhouette values, to study and understand why those profiles where far from the centre of their clusters.




```
##          Crime.Type Victim.Sex Victim.Age Perpetrator.Sex Perpetrator.Age Weapon
## 51663             1          0         27               0              18      1
## 57870             1          0         21               0              22      1
## 2986              1          0         22               1              25      1
## 29925             1          0         56               0              24      1
## 95246             1          0         16               0              16      1
## 68293             1          0         43               0              20      0
## 62555             1          0         23               0              26      0
## 45404             1          1         44               0              44      1
## 65161             1          0         40               0              56      1
## 46435             1          0         46               0              28      1
## 9642              1          0         52               0              26      0
## 59134             1          1         23               0              30      1
## 52132             1          0         24               0              22      1
## 96849             0          0          2               0              77      1
## 14183             1          0         19               0              20      0
## 15180             1          0         22               0              30      1
## 27168             1          0         37               0              23      1
## 89709             1          1         31               0              38      1
```

8. Hierarchical Clustering (Alternative Technique)

As an alternative to K-Means I applied hierarchical clustering on a random subset of 10,000 observations (to reduce computation time). The analysis was performed using only the first two principal components and calculating the Euclidean distance between data points and applying the Ward.D2 linkage method, which aims to minimize the total within-cluster variance at each step. The result is visualized through a dendrogram where I manually selected **3 clusters** as in the previous K-Means approach.



The comparison between K-Means and hierarchical clustering shows **strong consistency** between the two methods: majority of observations from each K-Means cluster are mapped to a corresponding hierarchical cluster. This overlap reinforces the **stability and reliability** of the clustering structure previously identified.

```
##              hc_clusters
## kmeans_sample    1    2    3
##             1 5399   22    0
##             2   46 1787    5
##             3  712  567 1462
```